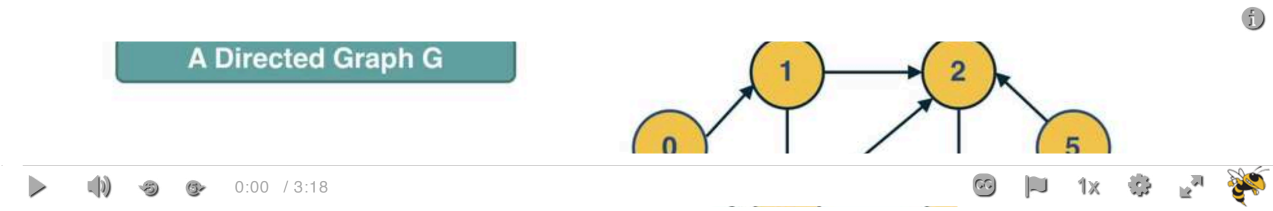
L2: Directed Acyclic Graphs (DAGs)



How to show the previous three properties

First, if a directed network has a topological order then it must be a DAG.

This is easy to show: if the network had a cycle, there would be an edge from a higher-rank node to a lower-rank node -- but this would violate the topological order property.

Second, a DAG must include at least one source node, i.e., a node with zero incoming edges. To see that, start from any node of the DAG and start moving backwards, following edges in the opposite direction. Given that there are no cycles and the graph has a finite number of nodes, we will eventually reach a source node.

Third, if a graph is a DAG then it must have a topological ordering.

You can show this as follows:

- Start from a source node s (we already showed that every DAG has at least one source node).
- Then remove that source node s and decrement the in-degree of all nodes that s points to. The graph is still a DAG after the removal of s.
- Choose a new source node s' and repeat the previous step until all nodes are removed.

Note that the topological order of a DAG may not be unique.